DISTINCT INFINITE CODES FORCE THE TREE SETS TO BE DEFINABLY INEQUIVALENT INSIDE DELTA_5

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Allowed standard prerequisites (editorial).

- The definable cardinality classes Delta_4 and Delta_5 and the relation of definable equivalence between two sets in their respective permutation models. Set up earlier in the paper and referred to in the counted statement.

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Fixes the comparison that is refuted and the conclusion that both sets lie in Delta_5 but not in Delta_4.

- The code of a weakly 2-transitive tree, recording the ramification orders along a maximal chain. Defined in the paragraph preceding the statement (source line 375). Supplies the invariant that the proof recovers from an assumed definable equivalence.
- Weak 2-transitivity and density of the maximal chains of T^+, giving automorphisms that move points inside the pieces U_a, L_ab and C_x while fixing a finite support. Used inside the counted proof at lines 382, 400-407 and 435-446. Lets the proof force incomparabilities and agreements of the two orders piecewise from the invariance of the support.

Theorem 0.1. *For any countable weakly 2-transitive trees T_1, T_2 in which all maximal chains of T_1^+, T_2^+ are dense, having distinct infinite codes, $|U_{T_1}|$ and $|U_{T_2}|$ are $\equiv$-inequivalent members of $\Delta_5 \setminus \Delta_4$.*

Proof. The point of the restriction we have made on our weakly 2-transitive trees is that for any ramification point r and $x, y > r$ in T, there is an automorphism of T fixing r and taking x to y.

Suppose for a contradiction that $|U_{T_1}| \equiv |U_{T_2}|$. In $\mathfrak{N}_{T_1}$ there must therefore be a structure $\prec$ on U_{T_1} such that $(U_{T_1}, \prec)$ in $\mathfrak{N}_{T_1}$ is elementarily equivalent to $(U_{T_2}, <)$ in $\mathfrak{N}_{T_2}$. Let A be a finite support for $(U_{T_1}, \prec)$ in $\mathfrak{N}_{T_1}$. Thus any automorphism of $(U_{T_1}, <)$ that fixes A pointwise also fixes $\prec$. We may assume that A is a subtree of U_{T_1}.

In order to handle this situation, we use the following notation, where A is a finite subset of T and $[A]$ is its definable closure, in this case just the lower subsemilattice generated by A. If a is maximal in A (or if $A = \emptyset$, a can be $-\infty$), we let $U_a = \{x \in T : a < x\}$, and if $a < b$ are consecutive members of $[A]$, then $L_{ab} = \{x \in T : a < x < b\}$ (the linear piece between a and b; again here a can be $-\infty$ in the case when b is minimal in A), and $S_{ab} = \{x \in T : a < x, b \not\leq x\}$ (the corresponding side piece). Now L_{ab} is linear, and we may also consider L_{ab}^+, which is the same thing, but calculated in T^+, that is, including ramification points too, which are coloured according to which orbit they lie in. If $x \in L_{ab}^+$, then $C_x = \{y \in T : b \wedge y = x\}$, which is the set of points which branch off from L_{ab}^+ at x. This is a union of cones at x.

A key remark is that the pointwise stabilizer G_A acts transitively on each U_a, L_{ab}, and C_x (here heavily using the assumption that the maximal chains in T^+ are dense). But more is true. In fact for any $b, c \in U_a$, there is an automorphism of T taking b to c which fixes A pointwise, *and* has support contained in U_a, with a similar statement holding for C_x. The analogous statement also holds for weak 2-transitivity. For instance, if $a < b < c$ and $a < d < e$, then by weak 2-transitivity there is an automorphism f fixing a and taking b to d, and applying weak 2-transitivity again, there is an automorphism g fixing d and taking $f(c)$ to e, and f, g may be chosen fixing all points below a and d respectively. Then gf fixes a and takes b to d and c to e.

Although not actually needed, we can describe the orbits of G_A. They are of the form U_a, and L_{ab}, as well as the union of the cones in S_{ab} which meet L_{ab} at points lying in some orbit of G_A on L_{ab}^+.

Now being a tree with dense chains is first order expressible, and so this must be true of $(U_{T_1}, \prec)$. We shall show that on each set of the form U_a or C_x, $\prec$ is equal to $<$ or the empty set (in which case it would be an antichain, i.e. pairwise $\prec$-incomparable). We just treat U_a, as C_x is similar.

First consider a maximal point a of A (and let a have a default value of $-\infty$ if $A = \emptyset$). First we note that if $b, c \in U_a$ are $<$-incomparable, then they are also $\prec$-incomparable. This is because there is an element of G_A swapping b and c. Since G_A preserves $\prec$, $b \prec c \Leftrightarrow c \prec b$, and since $\prec$ is antisymmetric, $\neg b \prec c \wedge \neg c \prec b$.

Next we show that there cannot be points $b, c \in U_a$ such that $b < c$ and $c \prec b$. For, if so, by weak 2-transitivity of G_A on U_a, for $x, y \in U_a$, $x < y \Rightarrow y \prec x$. Choose $z > a$ and incomparable $x, y > z$. Thus $x, y \prec z$. By the previous paragraph, x and y are also $\prec$-incomparable, so this contradicts $(U_{T_1}, \prec)$ a tree.

Next suppose that there are $b, c \in U_a$ such that $b < c$, and b and c are $\prec$-incomparable. By weak 2-transitivity of U_{T_1}, $a < x < y$ implies that x, y are $\prec$-incomparable. But also as shown above, if b and c are $<$-incomparable, they are also $\prec$-incomparable, and we deduce that $\prec$ is the empty relation on U_a.

Otherwise, it follows that for $x, y \in U_a$, $x < y \Rightarrow x \prec y$. Since also x, y $<$-incomparable $\Rightarrow$ x, y $\prec$-incomparable and $y < x \Rightarrow y \prec x$, we deduce that $x < y \Leftrightarrow x \prec y$. In other words, $<$ and $\prec$ agree on U_a.

Now let us consider S_{ab} where $a < b$ are consecutive members of A. First note that by a proof similar to the above, if x, y are $<$-incomparable members of C_c for $a < c < b$, then they are also $\prec$-incomparable, since there is a member of G_A which swaps x and y (and fixes c). We can deduce that if $a < c < d < b$, and $x \in C_c$, $y \in C_d$, then x and y are $\prec$-incomparable. For suppose that $x \prec y$ (a similar argument applying if $y \prec x$). Choose $x' \in C_c$ which is $<$-incomparable with x. Then by what we have just shown, x, x' are $\prec$-incomparable. Furthermore, there is a member of G_A which swaps x and x', and having support contained in C_c, and so this also fixes y. Since G_A preserves $\prec$, $x' \prec y$, but this contradicts $\prec$ a tree.

It follows that any $\prec$-comparabilities in S_{ab} must hold between members of the same C_c. Suppose then that $x < y$ and $x \prec y$ hold for some $x, y \in C_c$. As G_A acts weakly 2-transitively on C_c, $x < y \Rightarrow x \prec y$. Since we have also shown that x, y $<$-incomparable implies that they are $\prec$-incomparable, we deduce that $<$ and $\prec$ agree on C_c.

The main remaining point is to show that there is some a or c such that $\prec$ is non-empty on U_a or C_c. Suppose not. Then each U_a is an antichain, and since the members of distinct C_cs for $a < c < b$ are incomparable, each S_{ab} is also an antichain. Consider the possible relationship between $a \in A$ and members of U_b under $\prec$. The group G_A acts transitively on U_b, and so if any member of U_b is $\prec$-less than a, they all are, which violates $\prec$ a tree. Similarly, for members of C_c. The same argument applies to the possible relationship between members of distinct U_as, or C_cs, or between members of U_a and C_c. We deduce that the union of all the U_as and S_{ab}s is an antichain, and all its members are either incomparable with every member of A, or above some member of a. However, since $(U_{T_1}, \prec)$ is a tree satisfying the same first order sentences as $(U_{T_1}, <)$, all its chains are infinite, so this is impossible.

We deduce that there is a or c such that $<$ and $\prec$ agree on U_a or C_c. We just treat the first case, and show that for any $b \in U_a$, for $x \in T_1$, $b < x \Leftrightarrow b \prec x$. From left to right is already known. Suppose for a contradiction that there are some $b \in U_a$ and x such that $b \prec x$ but not $b < x$. Since $<$ and $\prec$ agree on U_a, $x \notin U_a$. Choose $c \in U_a$ incomparable with b. Then there is a member of G_A taking b to c and with support contained in U_a. This preserves $\prec$, and fixes x, and hence $c \prec x$. Since $\prec$ is a tree, b and c are $\prec$-comparable, and as they both lie in U_a, they are also $<$-comparable, which gives a contradiction.

Finally we choose any $b \in U_a$, and observe that the ramification orders arising above b in $<$ and $\prec$ are equal to those arising in T_1 and T_2 respectively. Since these

are determined by first order formulae, it follows that $|U_{T_1}|$ and $|U_{T_2}|$ have equal codes after all. □